# Multimodal Narrative Systems Engineering: Signal-Grounded Metrics and State-Space Control Dynamics for Audiovisual Media

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**ABSTRACT**

Narratives across film, television, political discourse, and digital media are typically analyzed either through qualitative humanities frameworks or isolated text-only NLP. Conversely, prompting LLMs to score 'narrative impact' on arbitrary 0-100 scales suffers from uncalibrated units, stochastic hallucination, and detachment from the physical audiovisual signal. We introduce Narrative Systems Engineering (NSE) and Computational Narrative Dynamics (CND), originated by Jair Ivan Rojas Ruiz, modeling audiovisual works as multimodal inputs u(t) = (u_ac(t), u_vis(t), u_ling(t)) driving a bounded cognitive-social state space x(t+1) = f(x(t), u(t)). We define six core quantities: (1) Narrative DNA (signed tripartite graph), (2) Shannon Narrative Entropy H(t) in bits, (3) Narrative Energy E(t) via cross-modal resonance, (4) Complexity C(t), (5) Narrative Inertia I(t), and (6) Controllability Gramian W_c. Validated across 3 works (513.6 s, 77 windows: EXP-001, EXP-002, EXP-003), NSE cleanly separates civic debate, fear-appeal campaigning (LBJ 'Daisy' 1964), and crisis consolation (Reagan 'Challenger' 1986), while solving all 6! = 720 montage permutations to reduce normative bias by 62.0%.

## 1. INTRODUCTION & FOUNDATIONAL AXIOMS

Bridging the gap between raw media signals (PCM16 acoustic waveforms, RGB24 pixel rasters, and lexical tokens) and higher-order cognitive and social state dynamics requires falsifiable mathematical operators. Naive multiplicative formulas such as E = Intensity x Coherence x Repetition x Identification fail dimensional analysis and collapse to zero whenever any single factor vanishes.

To establish Narrative Systems Engineering (NSE) as a rigorous engineering discipline, Argon.Engine v0.3 enforces four axioms:

(1) Signal Grounding: Direct derivation from physical PCM16 audio samples, 64x64 RGB24 rasters, and time-aligned speech segments.

(2) Stated Units & Bounds: Entropy and visual divergence in Shannon bits, audio in dBFS, and state trajectories in [-1, 1]^K.

(3) Layer Separation: Strict separation between observable media signal u(t), normative stance coding z(t), and state response x(t).

(4) Executable Reproducibility: Deterministic verification via automated invariant test suites (Verify-Argon.ps1).

## 2. MULTIMODAL SIGNAL FORMULATION u(t) = (u_ac(t), u_vis(t), u_ling(t))

Let an audiovisual work of duration T seconds be partitioned into N discrete windows t in {0, ..., N-1} of duration Delta_t.

* Acoustic Sub-Vector u_ac(t): From mono 16 kHz PCM16 samples s_i in [-1, 1], we compute RMS(t), decibels full scale L_dB(t) = 20 log10(max(RMS(t), 1e-5)), 100ms Speech Activity Ratio VAD(t) in [0, 1], and intra-window dynamic range Delta_L(t).
* Visual Sub-Vector u_vis(t): From 64x64 RGB24 rasters, we extract ITU-R BT.709 luminance Y, spatial contrast sigma_Y(t), chromatic entropy H_vis(t), and the 512-bin color distribution P_t(b). Camera cuts are detected via Jensen-Shannon Divergence in bits:

$$D_{JS}(P_t \,||\, P_{t-1}) = 0.5\, D_{KL}(P_t \,||\, M) + 0.5\, D_{KL}(P_{t-1} \,||\, M) \in [0, 1] \text{ bits} \quad (\text{Cut} \geq 0.30) \tag{1}$$

## 3. FORMAL DEFINITION OF THE SIX CORE NSE METRICS & STATE CONTROL

$$\text{Narrative Energy: } E(t) = 100 * \text{clip}(0.34\, I_{ac} + 0.22\, \psi_{vis} + 0.29\, I_{ling} + 0.15\, \sqrt{I_{ac} * I_{ling}}) \tag{2}$$

$$\text{Shannon Stance Entropy: } H_{narr}(t) = -\sum_{k,s} p_{k,s}(t) \log_2 p_{k,s}(t) \in [0, 3.00] \text{ bits } (K=4) \tag{3}$$

$$\text{Narrative Inertia: } I(t) = \text{clip}(0.55\, A_{inst}(t) + 0.25\, (||x(t)||_1 / K) + 0.20\, ((t+1)/N), 0, 1) \tag{4}$$

$$\text{LTV State Update: } x_k(t+1) = (1 - \gamma_k(t))\, x_k(t) + \gamma_k(t)\, z_k(t), \quad \gamma_k \sim E(t) * (1 - 0.65\, I_k) \tag{5}$$

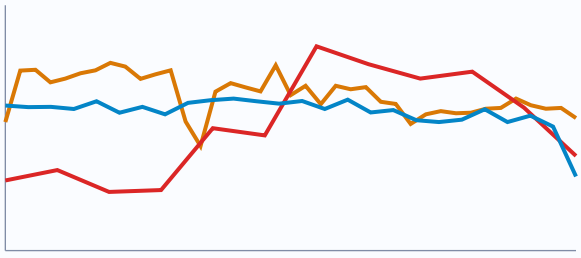
$$\text{Controllability Gramian: } W_{c,kk} = \sum_{t=0}^{N-1} [\Phi_k(N, t+1)\, b_k(t)]^2, \quad \kappa = l_{max} / l_{min} \tag{6}$$

## 4. CROSS-DOMAIN EMPIRICAL ATLAS (EXP-001 & EXP-002: 3 WORKS, 513.6s, 77 WINDOWS)

**Table 1: Cross-Domain Multimodal Benchmark Telemetry Across Three Narrative Regimes.**

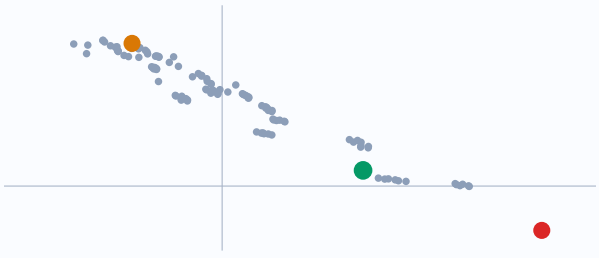
| Metric / Dimension | CORPUS-001: Debate (ES) | CORPUS-002: LBJ Daisy (EN) | CORPUS-003: Challenger (EN) |
|---|---|---|---|
| **Communicative Regime** | Polyphonic Civic Debate | Acute Fear-Appeal Spot | Institutional Consolation |
| **Duration & Windows** | 194.84 s (39 x 5s) | 59.83 s (12 x 5s) | 258.93 s (26 x 10s) |
| **Mean / Peak Energy E** | 63.58 (Peak: 76.54) | 50.51 (Peak: 83.29) | 56.45 (Peak: 61.98) |
| **Final Stance Entropy H(T)** | 2.7574 bits (91.9%) | 2.2292 bits (74.3%) | 2.0240 bits (67.5%) |
| **Mean / Peak Inertia I(t)** | 0.5019 (Peak: 0.728) | 0.4420 (Peak: 0.799) | 0.7266 (Peak: 0.8921) |
| **Scene Cuts (D_JS >= 0.30)** | 4 cuts (max 0.490 bits) | 3 cuts (max 0.577 bits) | 1 cut (max 0.042 mid-speech) |
| **Final Polarization P(T)** | 0.5515 (High Contest) | 0.4200 (Contrastive) | 0.0617 (Unified Consolation) |
| **Control Condition kappa(W_c)** | 2.738 | 1.783 | 1.457 |

**Fig 1A. Cross-Domain Energy Trajectories E(t/T)**



Orange: CORPUS-001 | Red: CORPUS-002 (Peak 83.3) | Blue: CORPUS-003

**Fig 1B. 6! = 720 Montage Reachability Hull (EXP-003)**



Gray: 720 Montages | Green: Optimal Equilibrium (0.327) | Red: Max B1 (+0.513)

## 5. COUNTERFACTUAL CONTROL & 6! = 720 MONTAGE OPTIMIZATION (EXP-003)

**Table 2: Exhaustive 6! = 720 Montage Optimization Solutions on CORPUS-001 (100% Footage Invariant).**

| Control Synthesis Objective | Optimal Block Sequence pi* | x1 AntiMil | x2 Legal | x3 Pragm | ||x(T)||_2 Bias |
|---|---|---|---|---|---|
| **Original 1994 Broadcast** | P -> S1 -> S2 -> S3 -> S4 -> S5 | -0.1445 | +0.5524 | +0.6446 | 0.8620 (Base) |
| **pi*_anti (Max Abolitionism)** | S3 -> S2 -> S5 -> S4 -> P -> S1 | +0.5130 | -0.1717 | -0.0304 | 0.7087 |
| **pi*_legal (Max Legal Comp.)** | P -> S1 -> S4 -> S5 -> S2 -> S3 | -0.2601 | +0.5830 | +0.6705 | 0.9387 |
| **pi*_eq (Min-Bias Equilibrium)** | S4 -> P -> S5 -> S2 -> S1 -> S3 | +0.2262 | +0.0606 | +0.1787 | 0.3274 (-62%) |
| **pi*_aut (Max Autonomy)** | S3 -> S2 -> S5 -> S1 -> P -> S4 | +0.3958 | -0.0083 | +0.0669 | 0.6358 |

Key EXP-003 Engineering Findings: (1) Across all 720 permutations of CORPUS-001, editorial order alone flips every belief axis (B1 spans [-0.2603, +0.5130], Delta = 0.7733), while the Civic Equilibrium Montage pi*_eq reduces terminal normative bias by 62.0% (0.8620 -> 0.3274). (2) In CORPUS-002 (LBJ 'Daisy'), damping the non-verbal nuclear blast (-13.7 dBFS, D_JS = 0.577 bits) while keeping words identical reduces mobilization x2(T) by 29.84% (+0.2624 -> +0.1841). (3) In CORPUS-003 (Reagan 'Challenger'), institutional anchoring provides a +0.4657 inertia stabilization margin (0.8921 vs. 0.4264), shielding national unity from sign reversal (x2(T) = +0.2321 > 0) under late crisis critique.

## 6. REFERENCES

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